

## SAFETY DATA SHEET

Creation Date 23-Mar-2012

Revision Date 29-Mar-2024

Revision Number 5

### 1. Identification

**Product Name** 4-Chlorobenzotrifluoride

**Cat No. :** A15154

**CAS No** 98-56-6  
**Synonyms** 4-Chloro-alpha,alpha,alpha-trifluorotoluene; PCBTF

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids  
Skin Sensitization

Category 3  
Category 1B

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Flammable liquid and vapor  
May cause an allergic skin reaction

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                       | CAS No  | Weight % |
|---------------------------------|---------|----------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                                                               |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                                                                                                                                                             |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle |

**Notes to Physician**                      pain or flushing  
Treat symptomatically

## 5. Fire-fighting measures

|                                         |                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                       |
| <b>Flash Point</b>                      | 43 °C / 109.4 °F                                                                                                               |
| <b>Method -</b>                         | CC (closed cup)                                                                                                                |
| <b>Autoignition Temperature</b>         | 650 °C / 1202 °F                                                                                                               |
| <b>Explosion Limits</b>                 |                                                                                                                                |
| <b>Upper</b>                            | No data available                                                                                                              |
| <b>Lower</b>                            | No data available                                                                                                              |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                       |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Gaseous hydrogen fluoride (HF). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system.                                                                                                                     |
| <b>Methods for Containment and Clean Up</b> | Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents.                                                            |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                       | ACGIH TLV                  | OSHA PEL                             | NIOSH                       | Mexico OEL (TWA)           |
|---------------------------------|----------------------------|--------------------------------------|-----------------------------|----------------------------|
| p-Chloro-a,a,a-trifluorotoluene | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                            |
|----------------------------------------|--------------------------------------------|
| Physical State                         | Liquid                                     |
| Appearance                             | Colorless                                  |
| Odor                                   | Fishy                                      |
| Odor Threshold                         | No information available                   |
| pH                                     | No information available                   |
| Melting Point/Range                    | -36 °C / -32.8 °F                          |
| Boiling Point/Range                    | 136 - 138 °C / 276.8 - 280.4 °F @ 760 mmHg |
| Flash Point                            | 43 °C / 109.4 °F                           |
| Method -                               | CC (closed cup)                            |
| Evaporation Rate                       | No information available                   |
| Flammability (solid,gas)               | Not applicable                             |
| Flammability or explosive limits       |                                            |
| Upper                                  | No data available                          |
| Lower                                  | No data available                          |
| Vapor Pressure                         | 38.6 mbar @ 50 °C                          |
| Vapor Density                          | 6.23                                       |
| Specific Gravity                       | 1.350                                      |
| Solubility                             | 0.0338 g/L @ 20 °C                         |
| Partition coefficient; n-octanol/water | No data available                          |
| Autoignition Temperature               | 650 °C / 1202 °F                           |
| Decomposition Temperature              | No information available                   |
| Viscosity                              | 0.67 cP at 38 °C                           |
| Molecular Formula                      | C7 H4 Cl F3                                |
| Molecular Weight                       | 180.56                                     |

## 10. Stability and reactivity

|                                         |                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                     |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                |
| <b>Conditions to Avoid</b>              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.                       |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                                                                        |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Gaseous hydrogen fluoride (HF), Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                  |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                       | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation               |
|---------------------------------|---------------------------|------------------------------|-------------------------------|
| p-Chloro-a,a,a-trifluorotoluene | LD50 = 5546 mg/kg ( Rat ) | LD50 > 3300 mg/kg ( Rabbit ) | LC50 = 32.03 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Irritating to eyes, respiratory system and skin

**Sensitization** May cause an allergic skin reaction

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                       | CAS No  | IARC     | NTP        | ACGIH      | OSHA | Mexico     |
|---------------------------------|---------|----------|------------|------------|------|------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | Group 2B | Not listed | Not listed | X    | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                       | Freshwater Algae | Freshwater Fish                                                                                           | Microtox                                                                     | Water Flea                                      |
|---------------------------------|------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------|
| p-Chloro-a,a,a-trifluorotoluene | Not listed       | LC50 = 3 mg/L, 96h semi-static (Danio rerio)<br>LC50 = 11.5 - 15.8 mg/L, 48h static (Lepomis macrochirus) | EC50 = 11.1 mg/L 5 min<br>EC50 = 13.4 mg/L 15 min<br>EC50 = 14.3 mg/L 30 min | EC50 = 2 mg/L, 48 h semi-static (Daphnia magna) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** Not likely to bioaccumulate.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                       | log Pow |
|---------------------------------|---------|
| p-Chloro-a,a,a-trifluorotoluene | 3.7     |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

**DOT**

UN-No UN2234  
 Proper Shipping Name CHLOROBENZOTRIFLUORIDES  
 Hazard Class 3  
 Packing Group III

**TDG**

UN-No UN2234  
 Proper Shipping Name CHLOROBENZOTRIFLUORIDES  
 Hazard Class 3  
 Packing Group III

**IATA**

UN-No UN2234  
 Proper Shipping Name CHLOROBENZOTRIFLUORIDES  
 Hazard Class 3  
 Packing Group III

**IMDG/IMO**

UN-No UN2234  
 Proper Shipping Name CHLOROBENZOTRIFLUORIDES  
 Hazard Class 3  
 Packing Group III

### 15. Regulatory information

**United States of America Inventory**

| Component                       | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------------------|---------|------|-----------------------------------------------|-----------------------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | X    | ACTIVE                                        | -                           |

**Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                       | CAS No  | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|---------------------------------|---------|-----|-----|-----------|-------|------|------|------|-------|----------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | X   | -   | 202-681-1 | X     | X    | X    | X    | X     | KE-05928 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313****SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product contains the following Proposition 65 chemicals.

| Component                       | CAS No  | California Prop. 65 | Prop 65 NSRL | Category   |
|---------------------------------|---------|---------------------|--------------|------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | Carcinogen          | 23 µg/day    | Carcinogen |

**U.S. State Right-to-Know Regulations**

| Component                       | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------------------|---------------|------------|--------------|----------|--------------|
| p-Chloro-a,a,a-trifluorotoluene | -             | X          | -            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

## Mexico - Grade

No information available

## Authorisation/Restrictions according to EU REACH

Not applicable

| Component                       | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | -                                                                   | -                                                                             | -                                                                                                     |

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                       | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

See table for values

| Component                                        | OECD PFAS | US (EPA) PFAS | EU (ECHA) PFAS | UK (HSE) PFAS | Chemsec PFAS (Sin List) |
|--------------------------------------------------|-----------|---------------|----------------|---------------|-------------------------|
| p-Chloro-a,a,a-trifluorotoluene (CAS #: 98-56-6) | -         | -             | Listed         | Listed        | Listed                  |

## PFAS Legend

Listed = Meets the PFAS definition of the named authority

## Other International Regulations

| Component                       | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| p-Chloro-a,a,a-trifluorotoluene | 98-56-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y45                      |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

## Creation Date

23-Mar-2012

## Revision Date

29-Mar-2024

## Print Date

29-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS